



Anchor-guided global view reconstruction for multi-view multi-label feature selection

Pingting Hao^{a,b}, Kunpeng Liu^c, Wanfu Gao^{a,b,*}

^a College of Computer Science and Technology, Jilin University, China

^b Key Laboratory of Symbolic Computation and Knowledge Engineering of Ministry of Education, Jilin University, China

^c Department of Computer Science, Portland State University, Portland, OR 97201 USA

ARTICLE INFO

Keywords:

Multi-view learning
Multi-label learning
Feature selection
Reconstructed view

ABSTRACT

In multi-view multi-label learning (MVML), the accuracy of feature weights is pivotal for establishing feature order. However, conventional MVML methods often struggle with integrating distinct information from multiple views effectively, leading to unclear segmentation and potential noise introduction. To address this challenge, this paper proposes an anchor-based latent representation method for global view learning in MVML. Specifically, we encode inherent information from each view to derive a candidate multi-view representation. Anchors extracted from both the candidate view and the global view are then constrained to approximate equality in the latent space. Furthermore, a carefully designed view matrix serves as supplement, seamlessly integrated into the reconstruction process to augment information. The convergence of results is subsequently validated using multiplicative update rules. Experimental findings showcase the superior performance of our proposed method across various multi-view datasets.

1. Introduction

The objective of feature selection is twofold: to alleviate the computational burden induced by the “curse of dimensionality” and to bolster performance by pinpointing redundant and noisy features [1,2]. Its utility spans across unsupervised, single-label, and multi-label scenarios, contingent upon the complexity of relationships under consideration. Multi-view multi-label learning, anchored within the multi-label scenario, not only deals with the association of instances with multiple labels but also entails representing instances through multiple distinct views [3]. For instance, objects in an image can be observed from multiple viewpoints or angles, each offering a view with different features. Simultaneously, these objects are also identified by their respective labels. As the dimensionality of the feature space escalates, the complexity of relationships to be accounted for expands, transcending the dimensions themselves.

Multi-view multi-label learning distinguishes itself from traditional multi-label learning by aiming to harness a more comprehensive array of information. Its primary challenge lies in effectively balancing the incorporation of common and complementary information from diverse views. Current research efforts predominantly concentrate on establishing robust mappings between individual views and the label matrix. For instance, Liu et al. [4] introduce an objective function integrating both views and high-level latent semantic labels. These latent labels not only preserve the inherent relationships among instances but also bolster label correlation. Moreover, the label matrix associated with each view undergoes decomposition and refinement to capture its impact on classification performance, particularly in the field of ensemble methods [5]. Subsequently, feature weights are derived by analyzing

* Corresponding author at: College of Computer Science and Technology, Jilin University, China.

E-mail addresses: haopingting@jlu.edu.cn (P. Hao), kunpeng@pdx.edu (K. Liu), gaowf@jlu.edu.cn (W. Gao).

the interplay between individual views and view-specific labels. Nonetheless, feature weights derived solely from individual views necessitate additional constraints to handle scenarios involving overlapping or missing information.

Certain studies focus on learning a shared subspace that amalgamates different views to form a cohesive representation. Zhang et al. [6] introduce a regularizer that establishes a connection between the shared subspace and labels, under the assumption that multiple views can be amalgamated into a single view [7]. As delineated by the objective function in [6], each view is construed as a projection of the shared subspace, encapsulating a more diverse array of information compared to what is individually captured within each view [8]. Nonetheless, it remains imperative to explore the distinct characteristics of each view to effectively capture global-view information. Zhu et al. [9] devise mappings for the global-view and local-view, respectively, with the global-view synthesized by concatenating the original features. Regrettably, the oversight of noises and relationships within the feature space can impede the accurate estimation of feature weights. Consequently, the reconstruction of a global-view poses a challenge in the realm of MVML.

Given the imperative to discern complex relationships, anchor-based methods are well-suited to the requirements of multi-view multi-label learning. In contrast to directly combining each view, the anchor presents a small set of data points of landmarks after eliminating irrelevant components, thereby mitigating the interference caused by noise. Anchors are introduced primarily to alleviate computational burdens by substituting the full graph with an anchor graph. Furthermore, anchor-based techniques demonstrate robust adaptability to diverse data dimensions, facilitating accommodation of varying sizes of views [10]. Additionally, they are capable of capturing the latent relationship, thereby enhancing the ability of the model to comprehend complex data patterns [11]. Given these challenges and difficulties, we propose a novel anchor-based method named **Global-view Reconstruction with Anchor Guidance for Multi-view Multi-label Feature Selection (GRAFS)**. GRAFS aims to effectively reconstruct the global-view, which contains valuable comprehensive information. This reconstruction process involves leveraging two types of matrices: a purified shared subspace matrix as the candidate view and individual crafted view matrices, which contribute to the global-view information. By exploiting the approximate equality of two latent anchors, we guide the reconstruction process to achieve a high level of similarity between the candidate view and the reconstructed view. In comparison with existing methods, the main contributions of our method GRAFS can be summarized as follows:

- We propose a novel method for reconstructing the global-view by amalgamating information derived from both the shared subspace and distinct views. This facilitates the creation of an accurate mapping between the global-view and the label matrix.
- GRAFS employs regularization paradigms centered on exploring the information within the candidate view. By enforcing similarity between the two latent anchors, the method preserves the candidate view's information in the reconstructed view, thereby maintaining the integrity of shared information while incorporating distinctive information.
- To guarantee convergence, GRAFS utilizes multiplicative update rules in its optimization process. Experimental evaluations conducted on six prominent multi-view multi-label datasets validate the efficiency and outstanding performance of GRAFS.

In the rest of this paper, Section 2 presents a review of the existing literature. Section 3 offers a detailed explanation of our proposed method. Following this, Section 4 presents the experimental results and provides an in-depth analysis of the model. Finally, in Section 5, it concludes the paper by summarizing our findings and discussing potential directions for future research.

2. Related work

In this section, we provide a brief introduction to two related topics, namely, multi-label feature selection and multi-view multi-label learning.

2.1. Multi-label feature selection

In the definition of multi-label learning, each instance in the feature space is associated with a set of labels [12], distinguishing it from unsupervised learning. The classic works to this field, as demonstrated by Qian et al. [13], take into account various interrelationships, such as label correlation, feature correlation, and instance dependence, and consequently amalgamate these elements into a unified model to predict results. As a critical subset of multi-label learning and relevant to our study, the initial step in multi-label feature selection involves establishing a relationship between features and labels. This holds true regardless of whether the methods employed are categorized as filter-based, wrapper-based, or embedded-based.

- Filter-based methods, as exemplified in [14] and [15], typically utilize an importance assessment formula that integrates a term to compute the relevance between candidate features and labels.
- In certain wrapper-based methods [16,17], a heuristic function is employed to identify informative features with higher relevance to the labels and lower redundancy among themselves over multiple iterations.
- Similarly, embedded methods [18,19] also concentrate on constructing the mapping between features and labels through regularization paradigms within the objective function.

Specially for feature selection problem, the constraints on feature weights are of particular importance in multi-label scenario. Huang et al. [20] utilize the label correlation to constrain the structure of feature weight matrix, assuming that strongly correlated labels will have similar corresponding feature weights. Meanwhile, the properties of different norms are not consistent with each other. While the l_1 -norm promotes sparsity across all elements of the matrix, the $l_{2,1}$ -norm aims to achieve row-wise sparsity. To

combine the advantages of these norms, Li et al. [21] propose a method that selects label-specific and common features jointly. In addition to constraints on the complete feature weight matrix, Lin et al. [22] compute the positive or negative correlation matrix and incorporate it as a decomposition part of the feature weight matrix to enhance the relationship between features and labels. Moreover, Fan et al. [23] introduce a local discriminant model to constrain the feature correlation matrix, which is derived from the feature weight matrix, in order to capture the inherent structure within the feature space.

Indeed, the feature space in multi-view scenarios consists of multiple views rather than a unified space. Simply applying the aforementioned methods in their original form often fails to yield satisfactory outcomes. For comparable challenges that manifest in both multi-view and multi-label scenarios, the representation of heterogeneous data in a joint latent subspace and a discriminative latent space continue to pose significant challenges in the domain of multi-view feature learning [24]. Therefore, it becomes imperative to consider additional factors such as segmentation or reconstruction to address the complexities inherent in multi-view feature spaces.

2.2. Multi-view multi-label learning

With the increasing prevalence of multi-view scenarios, applications have expanded to encompass various domains, including but not limited to fake news detection, 3D object detection, and action recognition. As a result, numerous methods aim to strike a balance between consistency and complementarity among complex relationships, in order to achieve accurate and comprehensive information. Wu et al. [25] seek to augment inter-view communication by minimizing both confusion adversarial loss and multi-label loss, thereby facilitating shared subspace exploitation. Furthermore, they impose an orthogonal constraint for extracting information that is specific to each view. Zhu et al. [26] address this by constructing the structure of each latent space through leveraging the correlations with the projection of each view, label space and the Laplacian matrix of each view. Additionally, they ensure consistency between two views by maximizing a value measured based on the Hilbert-Schmidt independence criterion. Wang et al. [27] devise a method involving the construction of a non-linear self-representation term. They employ the concept of auto-encoders to constrain the affinity graph, thereby addressing the potential consistency and complementarity of data from multiple sources. Hao et al. [28] delve into the intricate relationships observed among variables within the splitting structure. This structure primarily comprises elements such as common labels, view-to-all specific labels, view-specific labels, and noisy labels. Pu et al. [29] classify various types of noises including slight perturbations, random corruptions, and outliers. To effectively handle these noises, they apply different norms as constraints, one of which is specifically designed based on the characteristics of outliers. Wang et al. [30] construct a consensus graph by calculating the dependency between the low-dimensional representation and the similarity matrix. They employ the Laplacian matrix as a means to eliminate the need for an additional clustering step. These findings reaffirm the existence of more complex relationships in multi-view multi-label (MVML) scenarios compared to traditional multi-label learning.

Recent research in the field of multi-view multi-label (MVML) learning has increasingly focused on addressing specific challenges encountered in real-world scenarios, including non-aligned scenarios and incomplete views. Huang et al. [31] develop an approach that involves the construction of a clustering indicator matrix and a similarity matrix through the application of matrix decomposition and self-expression paradigms. Then, they utilize graph Laplacian to integrate these two matrices, facilitating the learning of a comprehensive similarity matrix, even in missing scenario. Wang et al. [32] propose a method that incorporates not only available instances but also recovered instances. They leverage reverse projection to adeptly balance the discrepancies between the original data and the recovered data. Cai et al. [33] partition the complete feature set into two distinct components, known and learned missing matrices. Moreover, they employ the Laplacian matrix to preserve the similarity and consistency between pairs of instances. Tan et al. [34] address the issue of incomplete views and missing labels by devising a model that incorporates two indicator matrices to identify the locations of missing entries and labels. Similarly, Liu et al. [35] propose the implementation of a missing view indicator matrix along with an incomplete label indicator. These innovative introductions aim to mitigate the adverse impact during the computation of the multi-head self-attention score and the graph loss. Moreover, Li et al. [36] propose a non-aligned method that also tackles the challenges of incomplete views and missing labels. Notably, the regularizer paradigm in [36] is defined as a set of features corresponding to the same label to accommodate non-alignment. Despite these advancements, there are still areas within MVML that warrant further in-depth research.

3. The proposed method

In this section, we provide an overview of the proposed method in terms of problem formulation. Subsequently, we divide the detailed description of each part of GRAFS into two subsections: candidate view extraction and crafted view supplement. The objective function is presented in Section 3.4. Finally, we employ multiplicative rules to update each variable and analyze the complexity of the optimization algorithm.

3.1. Problem formulation

Given a dataset (X, Y) , X represents a set of different views denoted as $X^{(i)} \in \mathbb{R}^{n \times d(i)}$ where $i = 1, 2, \dots, V$, and V denotes the number of views. The observed label matrix for all views is denoted as $Y \in \{0, 1\}^{n \times l}$. The reconstructed view is represented as $X^f \in \mathbb{R}^{n \times d}$. The goal of this paper is to establish a mapping between the global-view and the label matrix, and the loss function is designed accordingly. The number of instances, labels and features are denoted by n , l and d . If the $x_m^{(i)} \in X^{(i)}$ is associated with the j -th label, then y_{mj} is set as 1. Additional notations can be found in Table 1.

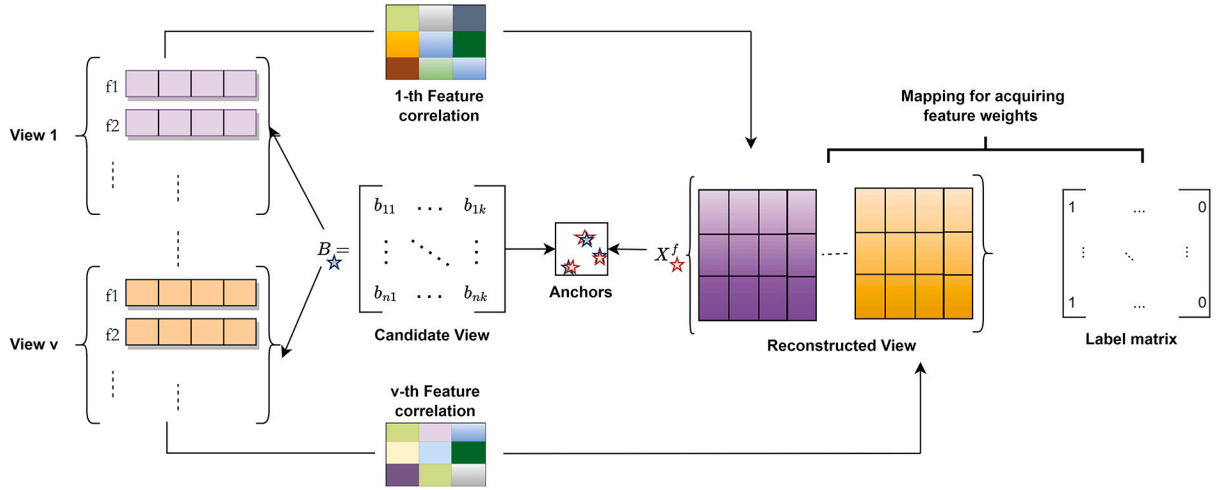


Fig. 1. An illustration of the proposed method GRAFS, which showcases the united procedure involving information extraction from the candidate view, information refilling from the crafted view, and the acquisition of feature weights.

Table 1
Description of Notations.

Symbol	Definition
k	The number of features for candidate view.
k_1	The number of clusters.
$d(i)$	The number of features for the i -th view.
$W \in \mathbb{R}^{d \times l}$	The feature weight matrix of global-view.
$B \in \mathbb{R}^{n \times k}$	The matrix for candidate view.
$P^{(i)} \in \mathbb{R}^{k \times d(i)}$	The projection coefficient matrix of the i -th view.
$W^c \in \mathbb{R}^{n \times k_1}$	The consensus cluster coefficient matrix.
$A^c \in \mathbb{R}^{k_1 \times k_1}$	The anchor of candidate view.
$A^f \in \mathbb{R}^{k_1 \times k_1}$	The anchor of reconstructed view.
$R^c \in \mathbb{R}^{k_1 \times k}$	The projection matrix of candidate view.
$R^f \in \mathbb{R}^{k_1 \times d}$	The projection matrix of reconstructed view.
$C^{(i)} \in \mathbb{R}^{d(i) \times d(i)}$	The contribution matrix for features in the i -th view.

Different from establishing separate mappings between each view and label, the challenge lies in how to reconstruct a holistic global view by effectively leveraging the information from each view $X^{(i)}$. This design aims to capture comprehensive information and simultaneously enhance feature selection. Basis on this, the objective of this paper is to develop a model that preserves the pure information for both common and distinctive features of each view. The objective function L_{GRAFS} consists three parts, including the feature selection L_F , extraction information L_E designed for the common part, and supplement L_S primarily aimed at integrating the distinctive part. The overall procedure of our proposed method GRAFS is described in Fig. 1.

$$L_{GRAFS} = L_F(\cdot) + L_E(\cdot) + L_S(\cdot). \quad (1)$$

3.2. Candidate view extraction

In this part, the common information from different views is represented as candidate view. We assume that a single latent representation can be transformed into multiple views through projection and other methods [8]. Instead of simply concatenating the views, we need to consider the relationships among them in our model. The construction procedure for the candidate view in our model takes into account both the view-relationship and the instance-relationship.

The view-relationship refers to the contribution of each view to the global-view matrix. To quantify this contribution, we consider the structural similarity between the local view and the global view as the measure of its impact.

A label matrix is utilized to encode global information, facilitating the assessment of consistency between the local structure similarity of each view and the global structure similarity [37]. This can be formulated as follows:

$$\frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n s_{ij}^{(v)} (y_{i.} - y_{j.})^2. \quad (2)$$

where $s_{ij}^{(v)}$ denotes the similarity between i -th instance and j -th instance in v -th view. The definition for $s_{ij}^{(v)}$ in the affinity matrix $S^{(v)}$ is formulated as:

$$s_{ij}^{(v)} = \begin{cases} \exp(-\frac{\|x_i^{(v)} - x_j^{(v)}\|_2^2}{\sigma^2}) & \text{if } x_i^{(v)} \in \mathcal{N}_q(x_j^{(v)}) \text{ or } x_j^{(v)} \in \mathcal{N}_q(x_i^{(v)}), \\ 0 & \text{otherwise} \end{cases}, \quad (3)$$

where σ represents a predefined parameter, and $\mathcal{N}_q(x_j^{(v)})$ represents the set of q nearest neighbors of $x_j^{(v)}$. Similarly, the meaning of $\mathcal{N}_q(x_i^{(v)})$ can be inferred. During the procedure of computation for similarity, we convert Formula (2) into the graph Laplacian representation [38]. The graph Laplacian matrix is denoted as $L_x^{(v)}$, where $L_x^{(v)} = A^{(v)} - S^{(v)}$. $A^{(v)}$ is the diagonal matrix, whose elements are $a_{ii}^{(v)} = \sum_{j=1}^n s_{ij}^{(v)}$. Hence, Formula (2) can be expressed in an alternative form as:

$$\text{Tr}(y^T (A^{(v)} - S^{(v)}) y) = \text{Tr}(y^T L_x^{(v)} y). \quad (4)$$

We assume non-negative view weights to represent the contribution of each view, with the constraint that the sum of view weights is equal to 1. The calculation formula is as follows:

$$v_i = \frac{1/\text{Tr}(y^T L_x^{(i)} y)}{\sum_{i=1}^V 1/\text{Tr}(y^T L_x^{(i)} y)}, \quad (5)$$

where v_i denotes the view weights of i -th view.

Next, the connection between a view and the candidate view B can be established through projection, where the projection matrix $P^{(i)}$ indicates the coefficient of instance-relationship between the two. By incorporating the view-relationship, we can construct a regularizer paradigm to obtain the candidate view as follows:

$$\min_{B, P^{(i)}} \sum_{i=1}^V v_i \|X^{(i)} - B P^{(i)}\|_F^2. \quad (6)$$

The main objective is to incorporate the information from the candidate view into the reconstructed view. In the process of solving for the candidate view, we aim to encode as much shared information from different views as possible into the latent representation, while simultaneously filtering out the noise present in the original views. However, a challenge arises due to the inconsistency in dimensions between the candidate view and the reconstructed view.

To leverage the advantages of the anchor, which offers flexibility across different dimensions and the ability to capture complex relationships, we decompose the matrix into three components. The first component ensures that the candidate view and reconstructed view share the same context, achieved through the use of a common cluster coefficient matrix W^c . The second component called anchor represents the latent information for the candidate view and the reconstructed view $A^{(c/f)}$, with the anchors of both views exhibiting a certain degree of similarity, despite not being identical. The final component represents the projection coefficient of the view $R^{(c/f)}$. Consequently, the extraction procedure for the reconstructed view can be summarized as follows:

$$\min_{X^f, B, W^c, A^c, R^c, A^f, R^f} \|B - W^c A^c R^c\|_F^2 + \|X^f - W^c A^f R^f\|_F^2 + \|A^c - A^f\|_F^2. \quad (7)$$

In summary, the process of extracting information from the candidate view involves two main steps, including the constructing procedure of candidate view and the extraction procedure. As a result, the design for the function of candidate view extraction is formulated as:

$$L_E(\cdot) = \sum_{i=1}^V v_i \|X^{(i)} - B P^{(i)}\|_F^2 + \|B - W^c A^c R^c\|_F^2 + \|X^f - W^c A^f R^f\|_F^2 + \|A^c - A^f\|_F^2. \quad (8)$$

3.3. Crafted view supplement

When it comes to the distinctive characteristic of MVML, it is necessary to fill in the missing partial characteristics of each view using the original view. To take into account the noise in the feature space, we construct an enhanced matrix by incorporating the contribution of features from the original view. Consequently, we impose a constraint on the similarity between the two matrices, aiming for them to be approximately close to each other. The regularizer paradigm between the reconstructed matrix and the enhanced matrix can be defined as follows, where $C^{(i)}$ denotes the contribution matrix in the i -th view:

$$\min_{G^{(i)}, C^{(i)}} \|G^{(i)} - X^{(i)} C^{(i)}\|_F^2. \quad (9)$$

Here, $G^{(i)}$ represents the portion obtained by splitting the reconstructed view. To extract the corresponding part based on the size of each view, we define the transformation matrix $D^{(i)} \in \mathbb{R}^{d \times d(i)}$. The size of $D^{(i)}$ depends on the splitting point of the global view. Assuming that the views are concatenated in order to form the reconstructed view, $D^{(i)}$ is a matrix with a diagonal offset of 1, where the offset is related to the number of features from previous views. For instance, if $d(i) = 2$ and $d = 3$, the matrix $D^{(i)}$ can be expressed as:

$$D^{(i)} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (10)$$

Therefore, Formula (9) is transformed into Formula (11) with the assistance of $D^{(i)}$. The regularizer paradigm is designed as:

$$\min_{X^f, C^{(i)}} \|X^f D^{(i)} - X^{(i)} C^{(i)}\|_F^2. \quad (11)$$

Similar to the term taken in the candidate view extraction, the view weights are incorporated into Formula (11) to account for the view relationship. Consequently, the function for the crafted view supplement is formulated as:

$$L_S(\cdot) = \sum_{i=1}^V v_i \|X^f D^{(i)} - X^{(i)} C^{(i)}\|_F^2. \quad (12)$$

3.4. The objective function

To address the feature selection problem, we derive the feature weight matrix for the global view by establishing a mapping between the reconstructed view and the labels. By computing the feature weight matrix, we can determine the order of feature and subsequently select features based on a specified number. Additionally, we leverage the benefits of the $l_{2,1}$ -norm to constrain the feature weight matrix, which exhibits group sparsity. This constraint ensures that the selected features are important for all labels.

$$L_F(\cdot) = \|X^f W - Y\|_F^2 + \|W\|_{2,1}. \quad (13)$$

By integrating the candidate view extraction and crafted view supplement, the objective function of GRAFS combines three functions within a unified framework. This objective function can be represented as:

$$\begin{aligned} L_{GRAFS} = & \|X^f W - Y\|_F^2 + \alpha \sum_{i=1}^V v_i \|X^{(i)} - B P^{(i)}\|_F^2 + \beta (\|B - W^c A^c R^c\|_F^2 \\ & + \|X^f - W^c A^f R^f\|_F^2 + \|A^c - A^f\|_F^2) \\ & + \gamma \sum_{i=1}^V v_i \|X^f D^{(i)} - X^{(i)} C^{(i)}\|_F^2 + \delta \|W\|_{2,1}, \end{aligned} \quad (14)$$

where α , β , γ and δ are trade-off parameters to ensure a balanced model. The primary variables associated with this objective function are B , $A^{c(f)}$ and $C^{(i)}$, which are benefit for the performance of reconstruction. The variable B is involved in two terms: one for constructing the candidate view and the other for factorizing it. Regarding the anchor, three terms utilize the variables to fit the reconstructed view. Enhancing the feature-relationship $C^{(i)}$ allows for obtaining accurate information from each view. Basis on this, the accuracy of the feature weights is improved based on the precision of the reconstructed view which contains the full perspective.

3.5. Optimization

The objective function of GRAFS exhibits non-smoothness and non-convexity. To address the non-smooth problem, we can relax the related norm, such as the $l_{2,1}$ -norm, as $\|W\|_{2,1} = 2Tr(W^T E W)$, where the diagonal matrix $e_{ii} = 1/(2\|W_i\|_2)$. Additionally, the equivalence formula for the Frobenius norm is defined as $\|Y\|_F^2 = Tr(Y^T Y)$. Regarding the non-convex problem, the variables in Formula (14) are solved separately in the optimization algorithm. In order to minimize the objective function, we utilize a non-negative alternating iterative optimization algorithm, which offers significant advantages in terms of non-parametric, non-negative, and stability. These three advantages are demonstrated by the elimination of the need to select hyperparameters such as learning rate during the solution process, the assurance that all matrix elements remain non-negative throughout the update process, and the observed stable convergence behavior. Consequently, the transformation of Formula (14) can be expressed as:

$$\begin{aligned} \min_{VS} Tr(X^f W - Y)^T (X^f W - Y) + \alpha \sum_{i=1}^V v_i Tr(X^{(i)} - B P^{(i)})^T (X^{(i)} - B P^{(i)}) \\ + \beta (Tr(B - W^c A^c R^c)^T (B - W^c A^c R^c) + Tr(X^f - W^c A^f R^f)^T \\ (X^f - W^c A^f R^f) + Tr(A^c - A^f)^T (A^c - A^f)) \\ + \gamma \sum_{i=1}^V v_i Tr(X^f D^{(i)} - X^{(i)} C^{(i)})^T (X^f D^{(i)} - X^{(i)} C^{(i)}) + 2\delta Tr(W^T E W) \\ - Tr(\omega X^f)^T - Tr(\psi W^T) - Tr(\rho B^T) - Tr(\rho P^{(i)T}) - Tr(\sigma W^c)^T \\ - Tr(\varsigma A^c)^T - Tr(\tau R^c)^T - Tr(v A^f)^T - Tr(\phi R^f)^T - Tr(\phi C^{(i)T}), \end{aligned} \quad (15)$$

where the variable set (VS) is $\{X^f, W, B, P^{(i)}, W^c, A^c, R^c, A^f, R^f, C^{(i)}\}$, then $\omega \in \mathbb{R}^{n \times d}$, $\psi \in \mathbb{R}^{d \times l}$, $\rho \in \mathbb{R}^{n \times k}$, $\phi \in \mathbb{R}^{k \times d(i)}$, $\sigma \in \mathbb{R}^{n \times k_1}$, $\varsigma \in \mathbb{R}^{k_1 \times k_1}$, $\tau \in \mathbb{R}^{k_1 \times k}$, $v \in \mathbb{R}^{k_1 \times k_1}$, $\phi \in \mathbb{R}^{k_1 \times d}$ and $\phi \in \mathbb{R}^{d(i) \times d(i)}$ are represented as Lagrangian multipliers. The optimization procedure is outlined in Algorithm 1.

We fix $\{X^f, B, P^{(i)}, W^c, A^c, R^c, A^f, R^f, C^{(i)}\}$ and update W . The sub-objective for W and its updating rule can be formulated as follows:

$$\Theta_W = \text{Tr}(X^f W - Y)^T (X^f W - Y) + 2\delta \text{Tr}(W^T E W) - \text{Tr}(\psi W^T). \quad (16)$$

Then, the partial derivatives of Θ with respect to W are shown as:

$$\frac{\partial \Theta}{\partial W} = 2(X^{fT} X^f W + \gamma E W - X^{fT} Y) - \psi. \quad (17)$$

The updating rule of W is written according to the KKT conditions as:

$$W \leftarrow W \circ \frac{X^{fT} Y}{X^{fT} X^f W + \delta E W}. \quad (18)$$

The process for updating other variables mirrors that of the variable W . Consequently, the procedure is not introduced into much detail here.

$$B \leftarrow B \circ \frac{\alpha \sum_{i=1}^V v_i X^{(i)} P^{(i)T} + \beta W^c A^c R^c}{\alpha \sum_{i=1}^V v_i B P^{(i)} P^{(i)T} + \beta B}, \quad (19)$$

$$P^{(i)} \leftarrow P^{(i)} \circ \frac{B^T X^{(i)}}{B^T B P^{(i)}}, \quad (20)$$

$$W^c \leftarrow W^c \circ \frac{B R^{cT} A^{cT} + X^f R^{fT} A^{fT}}{W^c A^c R^c R^{cT} A^{cT} + W^c A^f R^f R^{fT} A^{fT}}, \quad (21)$$

$$A^c \leftarrow A^c \circ \frac{W^{cT} B R^{cT} + A^f}{W^{cT} W^c A^c R^c R^{cT} + A^c}, \quad (22)$$

$$R^c \leftarrow R^c \circ \frac{A^{cT} W^{cT} B}{A^{cT} W^{cT} W^c A^c R^c}, \quad (23)$$

$$A^f \leftarrow A^f \circ \frac{W^{cT} X^f R^{fT} + A^c}{W^{cT} W^c A^f R^f R^{fT} + A^f}, \quad (24)$$

$$R^f \leftarrow R^f \circ \frac{A^{fT} W^{cT} X^f}{A^{fT} W^{cT} W^c A^f R^f}, \quad (25)$$

$$C^{(i)} \leftarrow C^{(i)} \circ \frac{X^{(i)T} X^f D^{(i)}}{X^{(i)T} X^{(i)} C^{(i)}}, \quad (26)$$

$$X^f \leftarrow X^f \circ \frac{Y W^T + \beta W^c A^f R^f + \gamma \sum_{i=1}^V v_i X^{(i)} C^{(i)} D^{(i)T}}{X^f W W^T + \beta X^f + \gamma \sum_{i=1}^V v_i X^f D^{(i)} D^{(i)T}}. \quad (27)$$

Algorithm 1 GRAFS.

Input: Data matrices $\{X^{(i)}\}_{i=1}^V$; Label matrix Y .

Parameter: Parameters α, β, γ and δ .

Output: Set of selected features.

- 1: Initialize $X^f, W, B, P^{(i)}, W^c, A^c, R^c, A^f, R^f$ and $C^{(i)}$;
 - 2: **repeat**
 - 3: Update the matrix W, B and $P^{(i)}$ in accordance with the formulation specified in Formula (18)-(20);
 - 4: Update the matrix W^c, A^c and R^c in accordance with the formulation specified in Formula (21)-(23);
 - 5: Update the matrix $A^f, R^f, C^{(i)}$ and X^f in accordance with the formulation specified in Formula (24)-(27);
 - 6: Update the objective function (15);
 - 7: **until** Convergence
 - 8: Derive the ordered feature sequence by performing computations on the weights, denoted as $\|W_{(j)}\|_2$ where $j = 1, 2, 3, \dots, d$.
-

3.6. Complexity analysis

In the optimization algorithm of GRAFS, ten variables are updated iteratively. The overall computational complexity of the algorithm can be determined by the variable with the highest complexity. Assuming the dimension of each view is d and $n \gg d \gg l \gg k_1, d \gg k \gg k_1$, the maximal complexity of $W, C^{(i)}$ and X^f are $O(d^2n)$. The complexity for B and $P^{(i)}$ are $O(ndk)$, and for W^c, A^f and R^f are $O(ndk_1)$. The remaining variables such as A^c and R^c bring a computational complexity of $O(nkk_1)$. To sum up, the maximal computational complexity of the whole training process is approximated as $O(d^2n)$ for each iteration.

Table 2

The detailed information for the datasets in our experiments.

Views	Yeast	SCENE	OBJECT	VOC07	MIRFlickr	3Sources
View-1(d_1)	GE(79)	CH(64)	CH(64)	DH(100)	DH(100)	BBC(1000)
View-2(d_2)	PP(24)	CM(225)	CM(225)	GIST(512)	GIST(512)	Reuters(1000)
View-3(d_3)		CORR(144)	CORR(144)	HH(100)	HH(100)	Guardian(1000)
View-4(d_4)		EDH(73)	EDH(73)			
View-5(d_5)		WT(128)	WT(128)			
Instances(n)	2417	4400	6047	3817	4053	169
Features(d)	103	634	634	712	712	3000
Labels(l)	14	33	31	20	38	6

Table 3Experimental results (mean $\pm$ std) in terms of AP and Coverage, where the 1st/2nd best results are shown in boldface/underline.

Dataset	GRAFS	M2LD	MSFS	SSFS	MoRE	MRDM	CLML
AP $\uparrow$							
yeast	0.6609 $\pm$ 0.014	0.6587 $\pm$ 0.015	0.6495 $\pm$ 0.012	0.6555 $\pm$ 0.014	0.6552 $\pm$ 0.016	<u>0.6595 $\pm$ 0.016</u>	0.6496 $\pm$ 0.012
SCENE	0.7977 $\pm$ 0.010	0.7792 $\pm$ 0.014	0.7887 $\pm$ 0.008	0.7774 $\pm$ 0.009	0.7877 $\pm$ 0.007	<u>0.7929 $\pm$ 0.010</u>	0.7870 $\pm$ 0.008
OBJECT	0.4987 $\pm$ 0.030	0.4535 $\pm$ 0.026	0.4631 $\pm$ 0.028	0.4387 $\pm$ 0.020	0.4727 $\pm$ 0.030	0.4599 $\pm$ 0.019	<u>0.4756 $\pm$ 0.022</u>
VOC07	0.5913 $\pm$ 0.012	0.5763 $\pm$ 0.009	0.5751 $\pm$ 0.006	0.5829 $\pm$ 0.007	0.5664 $\pm$ 0.016	<u>0.5836 $\pm$ 0.008</u>	0.5790 $\pm$ 0.011
MIRFlickr	0.6795 $\pm$ 0.011	0.6436 $\pm$ 0.013	0.6417 $\pm$ 0.010	<u>0.6719 $\pm$ 0.013</u>	0.6322 $\pm$ 0.011	0.6407 $\pm$ 0.008	0.6442 $\pm$ 0.007
3Sources	0.4718 $\pm$ 0.037	0.4368 $\pm$ 0.038	0.4252 $\pm$ 0.035	<u>0.4385 $\pm$ 0.046</u>	-	0.4196 $\pm$ 0.041	0.4232 $\pm$ 0.046
Coverage $\downarrow$							
yeast	0.6356 $\pm$ 0.018	0.6425 $\pm$ 0.017	0.6375 $\pm$ 0.021	0.6428 $\pm$ 0.018	<u>0.6357 $\pm$ 0.022</u>	0.6426 $\pm$ 0.023	0.6407 $\pm$ 0.019
SCENE	0.4169 $\pm$ 0.014	0.4416 $\pm$ 0.016	0.4271 $\pm$ 0.012	0.4481 $\pm$ 0.011	0.4295 $\pm$ 0.010	<u>0.4252 $\pm$ 0.017</u>	0.4308 $\pm$ 0.012
OBJECT	0.2890 $\pm$ 0.021	0.3142 $\pm$ 0.030	0.3005 $\pm$ 0.027	0.3247 $\pm$ 0.015	0.302 $\pm$ 0.030	0.309 $\pm$ 0.031	<u>0.298 $\pm$ 0.020</u>
VOC07	0.4253 $\pm$ 0.015	0.4547 $\pm$ 0.015	0.4484 $\pm$ 0.016	<u>0.4315 $\pm$ 0.016</u>	0.4763 $\pm$ 0.027	0.4531 $\pm$ 0.014	0.4569 $\pm$ 0.015
MIRFlickr	0.5784 $\pm$ 0.008	0.6069 $\pm$ 0.009	0.6173 $\pm$ 0.013	<u>0.5829 $\pm$ 0.013</u>	0.6397 $\pm$ 0.010	0.6128 $\pm$ 0.017	0.6191 $\pm$ 0.015
3Sources	0.6167 $\pm$ 0.032	<u>0.6397 $\pm$ 0.03</u>	0.6602 $\pm$ 0.037	0.6602 $\pm$ 0.055	-	0.6723 $\pm$ 0.042	0.6727 $\pm$ 0.046

4. Experiments

In this section, we offer a comprehensive description of the experiments, encompassing details on the datasets utilized, comparison methods employed, and evaluation metrics applied. The experimental analysis entails a comparison of results, coupled with an examination of the method itself. This examination comprises a sensitivity analysis of parameters and a convergence analysis. Furthermore, the effectiveness of each component is demonstrated through an ablation study.

The experiments are conducted on a Pycharm platform, using an Intel Core i7-8700 processor, Windows 10 Operating System, and 32 GB RAM.

4.1. Experimental setup

To validate the performance of our method in terms of feature selection and other characteristics, we utilize six well-known multi-view multi-label datasets. These datasets exhibit varying numbers of views, ranging from 2 to 5, as depicted in Table 2. The number of features present in these datasets (yeast, SCENE and OBJECT, VOC07, MIRFlickr, 3Sources) varies from a few dozen to several thousand, thereby allowing us to evaluate the performance under diverse circumstances.

In addition, we select six prominent methods as powerful competitors, including both the multi-view multi-label methods (M2LD [39], MSFS [40]) and multi-label feature selection methods (SSFS [41], MoRE [42], MRDM [43] and CLML [21]). These comparison methods share a common emphasis on the feature space.

The experimental results are obtained through a five-fold cross-validation process, employing a grid search within the range of $\{10^{-3}, 10^{-2}, \dots, 10^3\}$ for parameter tuning. The effectiveness of GRAFS is evaluated using four metrics: Average Precision (AP), Coverage, Hamming Loss (HL), and Ranking Loss (RL). It should be noted that, for all metrics except AP, a lower value indicates better performance. The results are reported as the mean value and standard deviation in this paper.

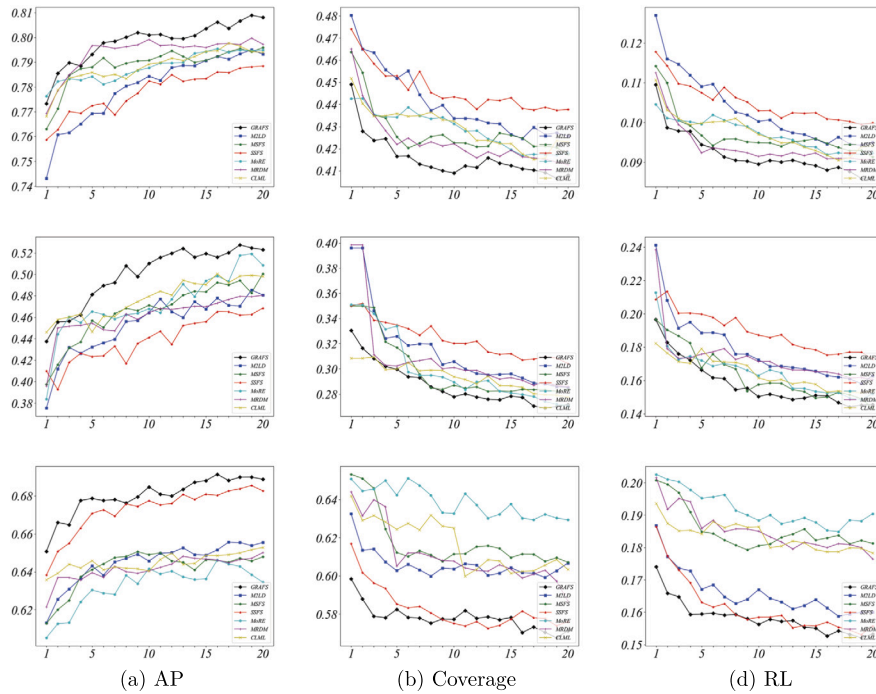
4.2. Experimental results

4.2.1. Feature selection performance

To objectively evaluate the performance of feature selection, we measure performance using varying numbers of features, ranging from 1% to 20% of the total number of features. The mean value for each metric is calculated by all sample points. The results of GRAFS and other comparison methods are presented in Tables 3–4, with the best performance highlighted in bold and the second best underlined. The performance trends are visually illustrated in Fig. 2. The following observations can be made:

Table 4Experimental results (mean $\pm$ std) in terms of HL and RL, where the 1st/2nd best results are shown in boldface/underline.

Dataset	GRAFS	M2LD	MSFS	SSFS	MoRE	MRDM	CLML
<i>HL $\downarrow$</i>							
yeast	0.2254 $\pm$ 0.003	0.2341 $\pm$ 0.002	0.2299 $\pm$ 0.004	0.2320 $\pm$ 0.004	<u>0.2259 $\pm$ 0.006</u>	0.2358 $\pm$ 0.007	0.231 $\pm$ 0.004
SCENE	0.0949 $\pm$ 0.005	0.1040 $\pm$ 0.006	0.0999 $\pm$ 0.005	0.1058 $\pm$ 0.005	0.1007 $\pm$ 0.005	<u>0.0979 $\pm$ 0.005</u>	0.1010 $\pm$ 0.005
OBJECT	0.0557 $\pm$ 0.003	0.0597 $\pm$ 0.003	0.0574 $\pm$ 0.003	0.0632 $\pm$ 0.003	0.0576 $\pm$ 0.003	0.0583 $\pm$ 0.003	<u>0.0573 $\pm$ 0.002</u>
VOC07	0.0858 $\pm$ 0.001	<u>0.0862 $\pm$ 0.002</u>	0.0893 $\pm$ 0.001	0.0886 $\pm$ 0.001	0.0890 $\pm$ 0.001	0.0880 $\pm$ 0.001	0.0884 $\pm$ 0.001
MIRFlickr	0.1741 $\pm$ 0.005	0.2029 $\pm$ 0.007	0.1944 $\pm$ 0.002	<u>0.1773 $\pm$ 0.008</u>	0.1931 $\pm$ 0.003	0.1946 $\pm$ 0.003	0.1917 $\pm$ 0.003
3Sources	0.2257 $\pm$ 0.016	0.2491 $\pm$ 0.025	<u>0.2357 $\pm$ 0.017</u>	0.2514 $\pm$ 0.021	-	0.2358 $\pm$ 0.017	0.2471 $\pm$ 0.018
<i>RL $\downarrow$</i>							
yeast	0.2547 $\pm$ 0.015	0.2605 $\pm$ 0.012	0.2614 $\pm$ 0.010	0.2593 $\pm$ 0.010	<u>0.2563 $\pm$ 0.017</u>	0.2574 $\pm$ 0.015	0.2628 $\pm$ 0.010
SCENE	0.0925 $\pm$ 0.007	0.1037 $\pm$ 0.009	0.0970 $\pm$ 0.006	0.1056 $\pm$ 0.005	0.0974 $\pm$ 0.005	<u>0.0949 $\pm$ 0.007</u>	0.0980 $\pm$ 0.006
OBJECT	0.1600 $\pm$ 0.017	0.1791 $\pm$ 0.019	<u>0.1648 $\pm$ 0.017</u>	0.1901 $\pm$ 0.012	0.1661 $\pm$ 0.016	0.1744 $\pm$ 0.017	0.1658 $\pm$ 0.013
VOC07	0.1977 $\pm$ 0.011	0.2142 $\pm$ 0.01	0.2133 $\pm$ 0.01	<u>0.2021 $\pm$ 0.01</u>	0.2298 $\pm$ 0.017	0.2124 $\pm$ 0.009	0.2157 $\pm$ 0.011
MIRFlickr	0.1586 $\pm$ 0.006	0.1672 $\pm$ 0.008	0.1855 $\pm$ 0.006	<u>0.1619 $\pm$ 0.009</u>	0.1925 $\pm$ 0.006	0.1858 $\pm$ 0.007	0.1844 $\pm$ 0.006
3Sources	0.4775 $\pm$ 0.037	<u>0.5045 $\pm$ 0.039</u>	0.5253 $\pm$ 0.043	0.5172 $\pm$ 0.060	-	0.5360 $\pm$ 0.051	0.5345 $\pm$ 0.055

**Fig. 2.** The performance for the feature selection is characterized on SCENE, OBJECT and MIRFlickr with three evaluation metrics.

- (1) Our method outperforms all baseline methods on all six datasets. Other methods, such as SSFS and MRDM, also exhibit strong performance, ranking second in approximately 29.2% and 25% of cases, respectively. SSFS exhibits a propensity to uncover the shared term by employing matrix factorization, thereby extracting latent feature information. Simultaneously, MRDM formulates a novel variable, imposing constraints on it by assessing the dependency between it and original label space. Hence, the strategic extraction of concealed information and the establishment of connections with pre-existing information emerge as crucial factors in enhancing performance.
- (2) While MoRE performs slightly worse than SSFS and MRDM, it emphasizes the utilization of information from the feature space and ignores other relationships. Conversely, CLML, which incorporates label correlation and instance correlation, exhibits similar performance of MoRE due to the lack of consideration for features in MVML.
- (3) Compared to multi-view multi-label methods, the non-recovery method (MSFS) and the method based on shared space (M2LD) demonstrate relatively poorer performance. Based on experimental findings, GRAFS is more likely to provide comprehensive information within the given constraints.

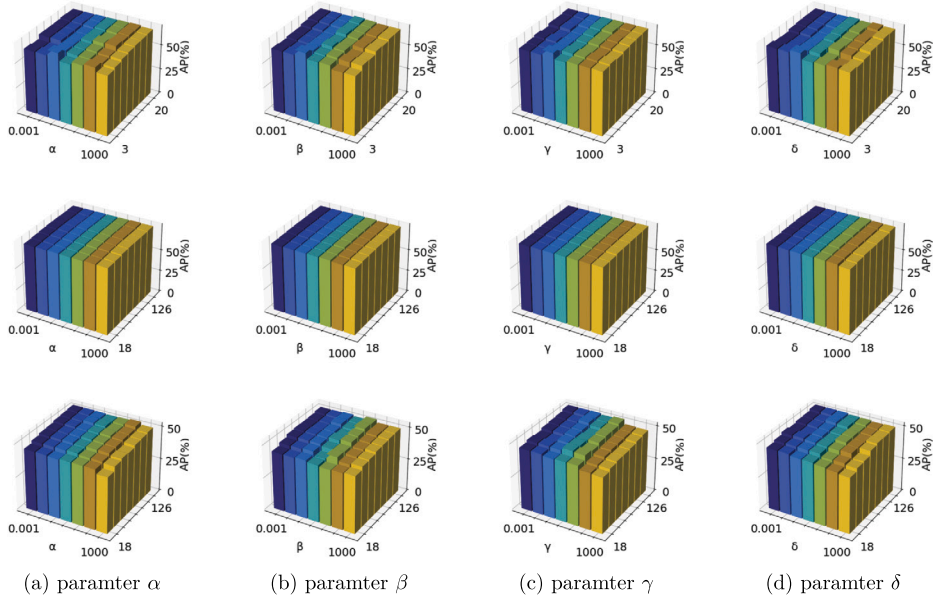


Fig. 3. Parameter sensitivity studies on the yeast, SCENE and OBJECT datasets.

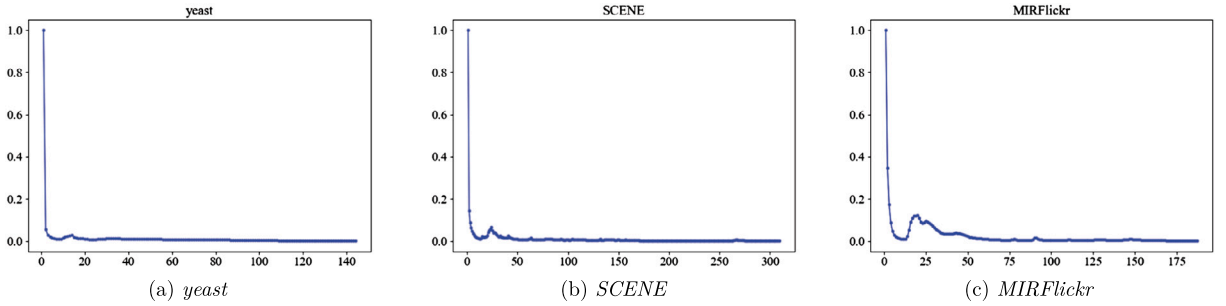


Fig. 4. Convergence curves on yeast, SCENE and MIRFlickr.

4.2.2. Parameter analysis

The sensitivity analysis of the four parameters is depicted in Fig. 3. The X-axis represents the range of grid search, spanning from 0.001 to 1000, while the Z-axis represents the AP metric value. It is evident that the performance remains stable across different numbers of features on the SCENE dataset. In contrast, some fluctuation is observed on the yeast and OBJECT datasets, but the magnitude of fluctuation is relatively small, especially when the number of instances is large. While the fluctuation patterns of the parameters differ, the maximum amplitudes on the yeast and OBJECT datasets do not exceed 3.6% and 7.0%, respectively.

4.2.3. Convergence

In Fig. 4, the curve illustrates the convergence trend and speed. The Y-axis represents the converging ratio $(Z^{t-1} - Z^t) / Z^{t-1}$, while the X-axis denotes the number of iterations. The curve demonstrates a positive and gradually decreasing speed, indicating the convergence towards stability. The speed is particularly fast at the beginning of the iteration. Before reaching approximate stability, some fluctuations occur, but they do not exceed 0.1. Overall, the method exhibits convergence and possesses a relatively rapid rate of convergence.

4.2.4. Running time comparison

Except the complexity of GRAFS, the complexity of M2LD, MSFS, SSFS, MoRE, MRDM and CLML is $O(k_2dn)$, $O(ndc + d^2 + n^2)$, $O(nda + n^2a)$, $O(n^2d + n^2\log(n))$, $O(n^3)$ and $O(nd^2 + d^2l + ndl)$ respectively. The symbol for k_2 represents the dimension of the latent subspace, the symbol for a is the number of already-selected features, and c denotes the number of clusters for labels. In Table 5, it can be seen that the MRDM actually has the slowest running time due to its complexity, only when the number of instances is small such as 3Sources. The complexity for CLML is similar with GRAFS, and MSFS has higher complexity when there is a significant difference between n and d .

Table 5
Running time (in second) of different methods.

Running time (sec)	GRAFS	M2LD	MSFS	SSFS	MoRE	MRDM	CLML
yeast	5.065	4.903	17.072	1.590	2.918	93.335	6.506
SCENE	167.411	14.337	120.570	9.042	43.108	856.762	91.750
OBJECT	86.887	25.802	178.429	18.098	73.001	2360.426	167.513
VOC07	73.400	16.212	33.903	7.340	39.242	181.571	70.510
MIRFlickr	98.878	17.956	26.707	10.729	42.756	219.514	98.398
3Sources	28.203	0.745	0.892	2.258	-	5.995	91.691

Table 6
Ablation experimental results of GRAFS in comparison with GRAFS_{ver1} and GRAFS_{ver2} on all datasets.

	Yeast	SCENE	OBJECT	VOC07	MIRFlickr	3Sources
GRAFS	0.6609	0.7977	0.4976	0.5913	0.6795	0.4718
GRAFS _{ver1}	0.6487	0.7959	0.4907	0.5764	0.6749	0.4596
GRAFS _{ver2}	0.6608	0.7977	0.4859	0.5856	0.6793	0.4687

4.2.5. Ablation study

To explore advantages of the design for GRAFS, the first issue is the dimension of the matrix B . It is necessary to discuss whether k is equal to d , resulting in the dimension change of B to $B \in \mathbb{R}^{n \times d}$. This change raises another concern that B can be seen as the reconstructed view, and the objective function, denoted as GRAFS_{ver1} is written as:

$$L_{GRAFS_{ver1}} = \|X^f W - Y\|_F^2 + \alpha \sum_{i=1}^V v_i \|X^{(i)} - X^f P^{(i)}\|_F^2 + \gamma \sum_{i=1}^V v_i \|X^f D^{(i)} - X^{(i)} C^{(i)}\|_F^2 + \delta \|W\|_{2,1}. \quad (28)$$

Compared to the original method, the objective function above is not only revised by replacing B with X^f , but also modifies the procedure for the extraction information from the candidate view. Another question pertains to the importance of the anchor. Specifically, it refers to the consistency of the compressed environment between two matrices, where A^c is equal to A^f . Consequently, GRAFS_{ver2} can be defined accordingly, and the results of comparison among these three types are presented in Table 6, computed based on the best performance obtained through grid searching.

$$L_{GRAFS_{ver2}} = \|X^f W - Y\|_F^2 + \alpha \sum_{i=1}^V v_i \|X^{(i)} - B P^{(i)}\|_F^2 + \beta (\|B - W^c R^c\|_F^2 + \|X^f - W^c R^f\|_F^2) + \gamma \sum_{i=1}^V v_i \|X^f D^{(i)} - X^{(i)} C^{(i)}\|_F^2 + \delta \|W\|_{2,1}. \quad (29)$$

Finally, it is necessary to include a distinctive part as a supplementary function in the objective function. To verify its effectiveness, the original setting of γ as 0 is employed, resulting in a modified objective function denoted as GRAFS_{ver3}. The results for comparison between GRAFS_{ver3} and the original are presented in Table 7, with all trade-off parameters set to 1.

Analysis of these tables reveals that GRAFS achieves impressive performance on the AP metric. Additionally, GRAFS_{ver2} generally outperforms GRAFS_{ver1}. During the process of compressing and expanding dimensions, the noise is filtered out and informative features are incorporated into the reconstructed view. The difference between GRAFS_{ver2} and the original is small, especially on yeast, SCENE and MIRFlickr, where two anchors may be nearly equal. The inherent attributes of these datasets imply a relatively distinct instance clustering. Even when there is a need to unearth latent information and conduct additional decomposition, the model continues to exhibit a robust fit to the data. However, GRAFS demonstrates superior performance on the other three datasets, highlighting the distinction between B and X^f .

Furthermore, the results in Table 7 confirm the impact of the distinctive part. What can be observed is a slight deficiency in the dissimilarity between GRAFS_{ver3} and the original when considering the SCENE dataset. This, coupled with the negligible variance between the original and GRAFS_{ver2}, implies that the utilization of shared information plays a significant role in reconstructing the feature space. When considering the OBJECT dataset, the feature contribution matrix, despite its commendable performance on other datasets, falls short in catering to the specific characteristics of the OBJECT dataset, thus highlighting the need for future improvements. Overall, each component makes an indispensable contribution to achieving the best performance of the model.

5. Conclusion

From a novel perspective, the proposed method GRAFS establishes a mapping between the reconstructed global-view and labels. It amalgamates information from the candidate view and the crafted view to fuse global-view information effectively. The procedure for extracting the candidate view involves encoding different views into a latent representation and constraining similar information

Table 7
Ablation experimental results of GRAFS in comparison with GRAFS_{ver3} on all datasets.

	Yeast	SCENE	OBJECT	VOC07	MIRFlickr	3Sources
GRAFS	0.6609	0.7975	0.4850	0.5912	0.6795	0.4718
GRAFS _{ver3}	0.6545	0.7973	0.4861	0.5887	0.6770	0.4552

with latent anchors. In the crafted view, the objective function considers the contribution of features. The experimental results validate the rationale behind GRAFS and demonstrate its superior performance in feature selection. Moreover, each regularizer paradigm in the objective function proves its effectiveness. Upon thorough analysis of the extensive experimental outcomes, it is posited that the efficacy of the model can be significantly enhanced by integrating optimization strategies specifically related to the label space. Furthermore, the universality of the feature contribution matrix can be substantially augmented through the incorporation of constraints, thereby fostering a more efficient process for view reconstruction.

CRedit authorship contribution statement

Pingting Hao: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Kunpeng Liu:** Writing – review & editing, Project administration, Formal analysis. **Wanfu Gao:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work is funded by: by Science Foundation of Jilin Province of China under Grant No. 20230508179RC, and China Postdoctoral Science Foundation funded project under Grant No. 2023M731281 and Changchun Science and Technology Bureau Project 23YQ05.

References

- [1] R. Zhang, F. Nie, X. Li, X. Wei, Feature selection with multi-view data: a survey, *Inf. Fusion* 50 (2019) 158–167.
- [2] W. Qian, J. Huang, F. Xu, W. Shu, W. Ding, A survey on multi-label feature selection from perspectives of label fusion, *Inf. Fusion* 100 (2023) 101948.
- [3] G. Lyu, X. Deng, Y. Wu, S. Feng, Beyond shared subspace: a view-specific fusion for multi-view multi-label learning, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, 2022, pp. 7647–7654.
- [4] B. Liu, W. Li, Y. Xiao, X. Chen, L. Liu, C. Liu, K. Wang, P. Sun, Multi-view multi-label learning with high-order label correlation, *Inf. Sci.* 624 (2023) 165–184.
- [5] D. Zhao, Q. Gao, Y. Lu, D. Sun, Learning view-specific labels and label-feature dependence maximization for multi-view multi-label classification, *Appl. Soft Comput.* 124 (2022) 109071.
- [6] C. Zhang, Z. Yu, Q. Hu, P. Zhu, X. Liu, X. Wang, Latent semantic aware multi-view multi-label classification, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, 2018.
- [7] M. White, X. Zhang, D. Schuurmans, Y.-I. Yu, Convex multi-view subspace learning, *Adv. Neural Inf. Process. Syst.* 25 (2012).
- [8] C. Zhang, H. Fu, Q. Hu, X. Cao, Y. Xie, D. Tao, D. Xu, Generalized latent multi-view subspace clustering, *IEEE Trans. Pattern Anal. Mach. Intell.* 42 (1) (2018) 86–99.
- [9] C. Zhu, D. Miao, Z. Wang, R. Zhou, L. Wei, X. Zhang, Global and local multi-view multi-label learning, *Neurocomputing* 371 (2020) 67–77.
- [10] Q. Ou, S. Wang, P. Zhang, S. Zhou, E. Zhu, Anchor-based multi-view subspace clustering with hierarchical feature descent, *Inf. Fusion* (2024) 102225.
- [11] J. Guo, J. Ye, Anchors bring ease: an embarrassingly simple approach to partial multi-view clustering, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, 2019, pp. 118–125.
- [12] M.-L. Zhang, Z.-H. Zhou, A review on multi-label learning algorithms, *IEEE Trans. Knowl. Data Eng.* 26 (8) (2013) 1819–1837.
- [13] K. Qian, X.-Y. Min, Y. Cheng, F. Min, Weight matrix sharing for multi-label learning, *Pattern Recognit.* 136 (2023) 109156.
- [14] J. Dai, W. Huang, C. Zhang, J. Liu, Multi-label feature selection by strongly relevant label gain and label mutual aid, *Pattern Recognit.* 145 (2024) 109945.
- [15] X.-A. Ma, W. Jiang, Y. Ling, B. Yang, Multi-label feature selection via maximum dynamic correlation change and minimum label redundancy, *Artif. Intell. Rev.* 56 (Suppl 3) (2023) 3099–3142.
- [16] M. Paniri, M.B. Dowlatabadi, H. Nezamabadi-Pour, Mlaco: A multi-label feature selection algorithm based on ant colony optimization, *Knowl.-Based Syst.* 192 (2020) 105285.
- [17] F. Karimi, M.B. Dowlatabadi, A. Hashemi, Semiaco: a semi-supervised feature selection based on ant colony optimization, *Expert Syst. Appl.* 214 (2023) 119130.
- [18] Y. Fan, J. Liu, J. Tang, P. Liu, Y. Lin, Y. Du, Learning correlation information for multi-label feature selection, *Pattern Recognit.* 145 (2024) 109899.
- [19] Z. Zhang, Z. Zhang, J. Yao, L. Liu, J. Li, G. Wu, X. Wu, Multi-label feature selection via adaptive label correlation estimation, *ACM Trans. Knowl. Discov. Data* (2023).
- [20] J. Huang, F. Qin, X. Zheng, Z. Cheng, Z. Yuan, W. Zhang, Q. Huang, Improving multi-label classification with missing labels by learning label-specific features, *Inf. Sci.* 492 (2019) 124–146.
- [21] J. Li, P. Li, X. Hu, K. Yu, Learning common and label-specific features for multi-label classification with correlation information, *Pattern Recognit.* 121 (2022) 108259.

- [22] Y. Lin, Z. He, L. Guo, W. Ding, Multi-label feature selection via positive or negative correlation, *IEEE Trans. Emerg. Top. Comput. Intell.* (2023).
- [23] Y. Fan, J. Liu, W. Weng, B. Chen, Y. Chen, S. Wu, Multi-label feature selection with local discriminant model and label correlations, *Neurocomputing* 442 (2021) 98–115.
- [24] G. Song, S. Wang, Q. Huang, Q. Tian, Learning feature representation and partial correlation for multimodal multi-label data, *IEEE Trans. Multimed.* 23 (2020) 1882–1894.
- [25] X. Wu, Q.-G. Chen, Y. Hu, D. Wang, X. Chang, X. Wang, M.-L. Zhang, Multi-view multi-label learning with view-specific information extraction, in: *IJCAI*, 2019, pp. 3884–3890.
- [26] P. Zhu, Q. Hu, Q. Hu, C. Zhang, Z. Feng, Multi-view label embedding, *Pattern Recognit.* 84 (2018) 126–135.
- [27] H. Wang, M. Yao, G. Jiang, Z. Mi, X. Fu, Graph-collaborated auto-encoder hashing for multiview binary clustering, *IEEE Trans. Neural Netw. Learn. Syst.* (2023).
- [28] P. Hao, K. Liu, W. Gao, Double-layer hybrid-label identification feature selection for multi-view multi-label learning, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, 2024, pp. 12295–12303.
- [29] X. Pu, H. Che, B. Pan, M.-F. Leung, S. Wen, Robust weighted low-rank tensor approximation for multiview clustering with mixed noise, *IEEE Trans. Comput. Soc. Syst.* (2023).
- [30] H. Wang, G. Jiang, J. Peng, R. Deng, X. Fu, Towards adaptive consensus graph: multi-view clustering via graph collaboration, *IEEE Trans. Multimed.* (2022).
- [31] Y. Huang, Z. Shen, Y. Cai, X. Yi, D. Wang, F. Lv, T. Li, Imufs: complementary and consensus learning-based incomplete multi-view unsupervised feature selection, *IEEE Trans. Knowl. Data Eng.* (2023).
- [32] H. Wang, M. Yao, Y. Chen, Y. Xu, H. Liu, W. Jia, X. Fu, Y. Wang, Manifold-based incomplete multi-view clustering via bi-consistency guidance, *IEEE Trans. Multimed.* (2024).
- [33] Y. Cai, H. Che, B. Pan, M.-F. Leung, C. Liu, S. Wen, Projected cross-view learning for unbalanced incomplete multi-view clustering, *Inf. Fusion* 105 (2024) 102245.
- [34] Q. Tan, G. Yu, C. Domeniconi, J. Wang, Z. Zhang, Incomplete multi-view weak-label learning, in: *Ijcai*, 2018, pp. 2703–2709.
- [35] C. Liu, J. Wen, X. Luo, Y. Xu, Incomplete multi-view multi-label learning via label-guided masked view- and category-aware transformers, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, 2023, pp. 8816–8824.
- [36] X. Li, S. Chen, A concise yet effective model for non-aligned incomplete multi-view and missing multi-label learning, *IEEE Trans. Pattern Anal. Mach. Intell.* 44 (10) (2021) 5918–5932.
- [37] X. Zhu, *Semi-Supervised Learning with Graphs*, Carnegie Mellon University, 2005.
- [38] F.R. Chung, *Spectral Graph Theory*, vol. 92, American Mathematical Soc., 1997.
- [39] W. Liu, J. Yuan, G. Lyu, S. Feng, Label driven latent subspace learning for multi-view multi-label classification, *Appl. Intell.* 53 (4) (2023) 3850–3863.
- [40] Y. Zhang, J. Wu, Z. Cai, S.Y. Philip, Multi-view multi-label learning with sparse feature selection for image annotation, *IEEE Trans. Multimed.* 22 (11) (2020) 2844–2857.
- [41] W. Gao, Y. Li, L. Hu, Multilabel feature selection with constrained latent structure shared term, *IEEE Trans. Neural Netw. Learn. Syst.* (2021).
- [42] S. Liu, X. Song, Z. Ma, E.D. Ganaa, X. Shen, More: multi-output residual embedding for multi-label classification, *Pattern Recognit.* 126 (2022) 108584.
- [43] R. Huang, Z. Wu, Multi-label feature selection via manifold regularization and dependence maximization, *Pattern Recognit.* 120 (2021) 108149.